



UNIVERSITI MALAYA

WIA2001 DATABASE PROJECT REPORT

TOPIC: FINAL REPORT

PREPARED BY

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DUE DATE

12/JAN/2026

1. Introduction

Raypro Tinted Expert is an outstanding automotive service provider. The company specialises in high-quality window tinting, ceramic coating, and Paint Protection Film (PPF) applications. Currently, the business operates through a traditional manual workflow where customers appointments are scheduled via WhatsApp and service details such as vehicle models, material choices, and customer contact information are recorded on paper or the boss' mobile devices. Under this manual system, financial transactions are documented with handwritten receipts, and warranty details are issued through physical cards.

As the complexity of the services grows, the business manages a polished pricing model that fluctuates based on vehicle size, material grades, and the specific area of coverage. For example, PPF services vary depending on whether a customer selects a partial, full front, or full car application. To maintain quality control, the center also mandates "before and after" photography and ensures that every job is assigned to at least one qualified technician.

To keep pace with increasing demand and the need for precision, Raypro Tinted Expert is moving toward a centralized digital management system. This transition aims to replace manual logging with a robust data architecture that integrates real-time inventory tracking for materials like film rolls and coating liquids, while providing a role-based access for Managers, Cashiers, and Technicians. By digitizing these operations, their business ensures that all vehicle records, payment histories, and warranty terms are preserved in a secure, searchable, and scalable environment.

2. Business rules

These are the business rules for RAYPRO TINTED EXPERT divided into three categories

A. Service & Pricing Structure

i. Variable Pricing Model

Service costs are calculated based on four key metrics: vehicle size, type of service, material grade/brand, and the duration of the warranty period.

ii. PPF Coverage Tiers

Paint Protection Film (PPF) pricing is specifically tiered based on the area of application:

1. Partial: Specific panels (e.g., hood or fenders).
2. Full Front: Complete frontal protection.
3. Full Car: Total exterior coverage.

B. Operational Standards & Documentation

i. Technical Staffing

To maintain quality standards, every service job must have at least one assigned technician.

- ii. Quality Assurance (QA) Photos
"Before and After" photography is mandatory for all projects. These images must be systematically archived using designated file paths for record-keeping and insurance purposes.
- iii. Warranty Management
Every completed job is digitally linked to a warranty record, which explicitly details the start date, duration, and legal terms.

C. Workflow & Administration

- i. Booking Policy
High-intensity services like Coating and PPF require a 1–3 day advance booking window; however, the facility remains open to walk-in clients based on bay availability.
- ii. Real-Time Inventory Tracking
To prevent stockouts, the inventory system must be updated immediately upon the use of film rolls, coating liquids, PPF sheets, and specialized tools.
- iii. Flexible Staffing Model
Personnel are assigned roles of Manager, Cashier, or Technician. To maximize operational efficiency, staff members may hold multiple designations (e.g., a Technician acting as a Cashier during peak times).

3. System Objectives

Centralized Management System

- Unified Data Architecture
Implement a single digital ecosystem to consolidate all customer profiles, vehicle histories, service records, job statuses, and payment transactions.
- Operational Efficiency
Optimize the search experience to provide staff with near-instant access to critical data, reducing administrative lag and improving customer response times.
- Advanced Resource Management
Ensure precise, real-time tracking of materials and stock levels to prevent shortage.
- Workforce Optimization
Facilitate the strategic assignment of technicians and provide an oversight view of current workshop workloads to prevent bottlenecks.

- **Data Integrity & Compliance**
Establish a secure, centralized repository for warranty information to ensure records are protected and easily retrievable for future claims.
- **Scalability for Growth**
Design the system with a multi-branch architecture to ensure seamless operational consistency and data synchronization as the business expands to new locations.

4. User requirements

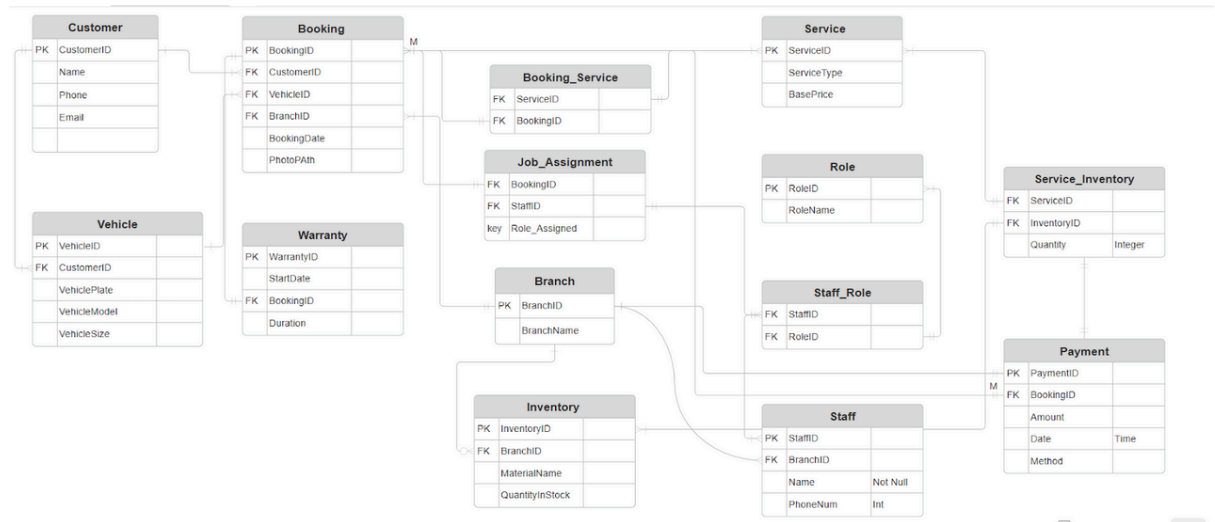
- A. General User (Administrative/View-Only)
 - a. Data Retrieval
Access to search all historical and active records.
 - b. Reporting
Capability to generate comprehensive analytics, including Sales Performance and Workload Distribution reports.
- B. Manager (Full Administrative Access)
 - a. System Oversight
Complete authority to view, edit, and manage all organizational data.
 - b. Strategic Control
Exclusive permissions to modify service pricing, update package details, and adjust business logic within the system.
- C. Cashier (Front-of-House Operations)
 - a. Customer Onboarding
Authorized to register new customer profiles and vehicle information.
 - b. Financial Transactions
Responsible for processing payments and generating official invoices for completed services.
- D. Technician (Service Execution)
 - a. Workflow Management
Access to a personalized dashboard to view daily job assignments and task priorities.
 - b. Resource Documentation
Real-time logging of materials and inventory consumed during the service process to ensure accurate stock tracking.

5. Entities and Attributes

Entity Name	Primary Key (PK)	Foreign Keys (FK)	Additional Attributes
Branch	BranchID	None	BranchName, Address
Staff	StaffID	BranchID	Name, PhoneNum
Role	RoleID	None	RoleName
Staff_Role	StaffID, RoleID (Composite)	StaffID, RoleID	None (Linker table for multiple roles)
Customer	CustomerID	None	Name, Phone, Email
Vehicle	VehicleID	CustomerID	VehiclePlate, VehicleModel, VehicleSize
Service	ServiceID	None	ServiceType, BasePrice
Inventory	InventoryID	BranchID	MaterialName, QuantityInStock
Service_Inventory	ServiceID, InventoryID	ServiceID, InventoryID	Quantity (Materials used per service)
Booking	BookingID	CustomerID, VehicleID, BranchID	BookingDate, PhotoPath
Booking_Service	BookingID, ServiceID	BookingID, ServiceID	None (Linker for services per booking)
Job_Assignment	BookingID, StaffID	BookingID, StaffID	RoleAssigned
Payment	PaymentID	BookingID, StaffID	Amount, Date, Method
Warranty	WarrantyID	BookingID	StartDate, Duration

6. ER Diagram

The following is the ER Diagram from the collected requirements. This diagram is a conceptual diagram only.



7. Normalization process

(1NF)

Customer(CustomerID,name,phone,email)

Vehicle(VehicleID,CustomerID,VehiclePlate,VehicleModel,VehicleSize)

Booking(BookingID,CustomerID,VehicleID,BranchID,BookingDate,,PhotoPath)

Warranty(WarrantyID,StartDate,BookingID,Duration)

Service(ServiceID,BasePrice,ServiceType)

Staff(StaffID,BranchID,Name,PhoneNum)

Role(RoleID(PK),RoleName)

Payment(PaymentID,BookingID,StaffID,Amount,Date,Method)

Inventory(InventoryID,MaterialName,BranchID,QuantityInStock)

Branch(BranchID,BranchName,Address)

Junction tables for M:N relationships

Staff_Role(StaffID,RoleName)

Booking_Service(ServiceID, BookingID)

Job_Assignment(BookingID,StaffID,RoleAssigned)

Service_Inventory(InventoryID,ServiceID,Quantity)

(2NF)

Customer(CustomerID,name,phone,email)

Vehicle(**VehicleID**,CustomerID(FK),VehiclePlate,VehicleModel,VehicleSize)
 Booking(**BookingID**,CustomerID(FK),VehicleID(FK),BranchID(FK),BookingDate,,PhotoPath)
 Warranty(**WarrantyID**,StartDate,BookingID(FK),Duration)
 Service(**ServiceID**,BasePrice,ServiceType)
 Staff(**StaffID**,BranchID(FK),Name,PhoneNum)
 Role(RoleID(PK),**RoleName**)
 Payment(**PaymentID**,BookingID(FK),StaffID(FK),Amount,Date,Method)
 Inventory(**InventoryID**,MaterialName,BranchID(FK),QuantityInStock)
 Branch(**BranchID**,BranchName,Address)

Junction tables for M:N relationships

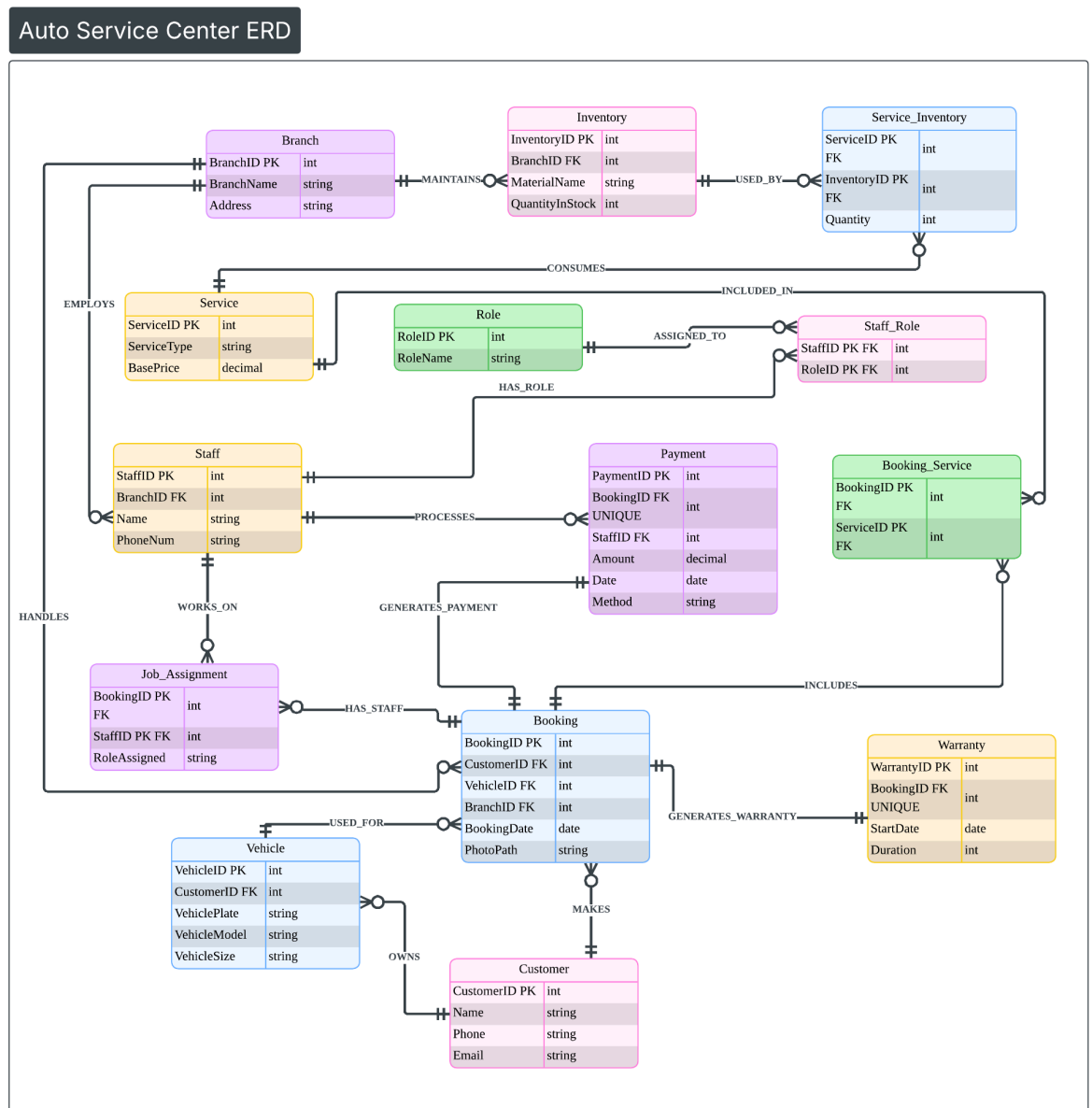
Staff_Role(**StaffID**(PK, FK),**RoleName**(PK, FK))
 Booking_Service(**ServiceID**(PK, FK), **BookingID**(PK, FK))
 Job_Assignment(**BookingID**(PK, FK),**StaffID**(PK, FK),RoleAssigned)
 Service_Inventory(**InventoryID**(PK, FK),**ServiceID**(PK, FK),Quantity)

(3NF)

Customer(**CustomerID**,name,phone,email)
 Vehicle(**VehicleID**,CustomerID(FK),VehiclePlate,VehicleModel,VehicleSize)
 Booking(**BookingID**,CustomerID(FK),VehicleID(FK),BranchID(FK),BookingDate,,PhotoPath)
 Warranty(**WarrantyID**,StartDate,BookingID(FK),Duration)
 Service(**ServiceID**,BasePrice,ServiceType)
 Booking_Service(**ServiceID**(PK, FK), **BookingID**(PK, FK))
 Staff(**StaffID**,BranchID(FK),Name,PhoneNum)
 Role(RoleID(PK),**RoleName**)
 Staff_Role(**StaffID**(PK, FK),**RoleName**(PK, FK))
 Job_Assignment(**BookingID**(PK, FK),**StaffID**(PK, FK),RoleAssigned)
 Inventory(**InventoryID**,MaterialName,BranchID(FK),QuantityInStock)
 Service_Inventory(**InventoryID**(PK, FK),**ServiceID**(PK, FK),Quantity)
 Payment(**PaymentID**,BookingID(FK),StaffID(FK),Amount,Date,Method)
 Branch(**BranchID**,BranchName,Address)

8. Implementable ERD

This diagram are the final diagram we implement inside the database



9. Sample SQL statements

CREATE TABLE Statements DDL:

-- BRANCH Table

```

CREATE TABLE BRANCH (
    BranchID NUMBER PRIMARY KEY,
    BranchName VARCHAR2(50) NOT NULL,
    Address VARCHAR2(100)
);
  
```

-- ROLE Table

```

CREATE TABLE ROLE (
  
```



```

        RoleID NUMBER PRIMARY KEY,
        RoleName VARCHAR2(30) NOT NULL
    );

-- STAFF Table
CREATE TABLE STAFF (
    StaffID NUMBER PRIMARY KEY,
    BranchID NUMBER REFERENCES BRANCH(BranchID),
    Name VARCHAR2(50) NOT NULL,
    PhoneNum VARCHAR2(15)
);

-- STAFF_ROLE (Bridge Table for Many-to-Many)
CREATE TABLE STAFF_ROLE (
    StaffID NUMBER REFERENCES STAFF(StaffID),
    RoleID NUMBER REFERENCES ROLE(RoleID),
    PRIMARY KEY (StaffID, RoleID)
);

-- CUSTOMER Table
CREATE TABLE CUSTOMER (
    CustomerID NUMBER PRIMARY KEY,
    Name VARCHAR2(50) NOT NULL,
    Phone VARCHAR2(15),
    Email VARCHAR2(50)
);

-- VEHICLE Table
CREATE TABLE VEHICLE (
    VehicleID NUMBER PRIMARY KEY,
    CustomerID NUMBER REFERENCES CUSTOMER(CustomerID),
    VehiclePlate VARCHAR2(15) NOT NULL,
    VehicleModel VARCHAR2(50),
    VehicleSize VARCHAR2(20)
);

-- SERVICE Table
CREATE TABLE SERVICE (
    ServiceID NUMBER PRIMARY KEY,
    ServiceType VARCHAR2(50) NOT NULL,
    BasePrice NUMBER
);

-- INVENTORY Table
CREATE TABLE INVENTORY (
    InventoryID NUMBER PRIMARY KEY,
    BranchID NUMBER REFERENCES BRANCH(BranchID),
    MaterialName VARCHAR2(50),

```

```

    QuantityInStock NUMBER
);

-- SERVICE_INVENTORY (Bridge Table)
CREATE TABLE SERVICE_INVENTORY (
    InventoryID NUMBER REFERENCES INVENTORY(InventoryID),
    ServiceID NUMBER REFERENCES SERVICE(ServiceID),
    Quantity NUMBER,
    PRIMARY KEY (InventoryID, ServiceID)
);

-- BOOKING Table
CREATE TABLE BOOKING (
    BookingID NUMBER PRIMARY KEY,
    CustomerID NUMBER REFERENCES CUSTOMER(CustomerID),
    VehicleID NUMBER REFERENCES VEHICLE(VehicleID)

```

Sample INSERT Statements DML:

```

-- Branches
INSERT INTO BRANCH VALUES (1, 'Downtown', '123 Main Street');
INSERT INTO BRANCH VALUES (2, 'Uptown', '456 High Street');

-- Roles
INSERT INTO ROLE VALUES (1, 'Manager');
INSERT INTO ROLE VALUES (2, 'Cashier');
INSERT INTO ROLE VALUES (3, 'Technician');

-- Staff
INSERT INTO STAFF VALUES (101, 1, 'Ali Ahmad', '0123456789');
INSERT INTO STAFF VALUES (102, 1, 'Siti Nur', '0198765432');
INSERT INTO STAFF VALUES (103, 2, 'John Tan', '0112233445');
INSERT INTO STAFF VALUES (104, 2, 'Sarah Lim', '0119988776');

-- Staff Roles
INSERT INTO STAFF_ROLE VALUES (101, 1); -- Ali is Manager
INSERT INTO STAFF_ROLE VALUES (101, 2); -- Ali also Cashier
INSERT INTO STAFF_ROLE VALUES (102, 3); -- Siti is Technician
INSERT INTO STAFF_ROLE VALUES (103, 3); -- John is Technician
INSERT INTO STAFF_ROLE VALUES (104, 2); -- Sarah is Cashier
INSERT INTO STAFF_ROLE VALUES (104, 3); -- Sarah also Technician

-- Customers
INSERT INTO CUSTOMER VALUES (201, 'Ahmad Faris', '0123344556',
'faris@example.com');
INSERT INTO CUSTOMER VALUES (202, 'Lina Wong', '0191122334',
'lina@example.com');

```

-- Vehicles

```
INSERT INTO VEHICLE VALUES (301, 201, 'WXY1234', 'Toyota Camry', 'Full');
INSERT INTO VEHICLE VALUES (302, 202, 'ABC5678', 'Honda Civic', 'Partial');
```

-- Services

```
INSERT INTO SERVICE VALUES (401, 'Window Tinting', 300);
INSERT INTO SERVICE VALUES (402, 'Ceramic Coating', 800);
INSERT INTO SERVICE VALUES (403, 'PPF Full Car', 2500);
```

-- Inventory

```
INSERT INTO INVENTORY VALUES (501, 1, 'PPF Sheet', 50);
INSERT INTO INVENTORY VALUES (502, 1, 'Ceramic Liquid', 20);
INSERT INTO INVENTORY VALUES (503, 2, 'Film Roll', 30);
```

-- Service Inventory

```
INSERT INTO SERVICE_INVENTORY VALUES (501, 403, 3); -- 3 sheets for PPF Full Car
INSERT INTO SERVICE_INVENTORY VALUES (502, 402, 1); -- 1 unit ceramic liquid
INSERT INTO SERVICE_INVENTORY VALUES (503, 401, 1); -- 1 roll film for Tinting
```

-- Bookings

```
INSERT INTO BOOKING VALUES (601, 201, 301, 1,
TO_DATE('2026-01-12','YYYY-MM-DD'), 'C:\Photos\001_before.jpg');
INSERT INTO BOOKING VALUES (602, 202, 302, 2,
TO_DATE('2026-01-13','YYYY-MM-DD'), 'C:\Photos\002_before.jpg');
```

-- Booking Service

```
INSERT INTO BOOKING_SERVICE VALUES (601, 401);
INSERT INTO BOOKING_SERVICE VALUES (601, 402);
INSERT INTO BOOKING_SERVICE VALUES (602, 403);
```

-- Job Assignment

```
INSERT INTO JOB_ASSIGNMENT VALUES (601, 102, 3); -- Siti as Technician
INSERT INTO JOB_ASSIGNMENT VALUES (601, 101, 2); -- Ali as Cashier
INSERT INTO JOB_ASSIGNMENT VALUES (602, 103, 3); -- John as Technician
INSERT INTO JOB_ASSIGNMENT VALUES (602, 104, 2); -- Sarah as Cashier
INSERT INTO JOB_ASSIGNMENT VALUES (602, 104, 3); -- Sarah also Technician
```

-- Payment

```
INSERT INTO PAYMENT VALUES (701, 601, 101, 1100,
TO_DATE('2026-01-12','YYYY-MM-DD'), 'Cash');
INSERT INTO PAYMENT VALUES (702, 602, 104, 2500,
TO_DATE('2026-01-13','YYYY-MM-DD'), 'Card');
```

-- Warranty

```
INSERT INTO WARRANTY VALUES (801, 601,
TO_DATE('2026-01-12','YYYY-MM-DD'), 12);
```

```
INSERT      INTO      WARRANTY      VALUES      (802,      602,  
TO_DATE('2026-01-13','YYYY-MM-DD'), 24);
```

Example Complex Query :

```
-- Workload Report: Who is working on which booking and role  
SELECT ba.BookingID, s.Name AS StaffName, r.RoleName  
FROM JOB_ASSIGNMENT ba  
JOIN STAFF s ON ba.StaffID = s.StaffID  
JOIN ROLE r ON ba.RoleAssigned = r.RoleID  
ORDER BY ba.BookingID, r.RoleName;
```

10. Screenshots of tables from the developed system

- DDL

APEX

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Search

RD

SQL Commands

Schema WKSP_DATABASE76

Language SQL

Rows 10

Clear Command

Find Tables

Save

Run

1 DESC Payment;

2

Results

Explain

Describe

Saved SQL

History

Object Type TABLE

Object PAYMENT

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
PAYMENT	PAYMENTID	NUMBER	22	-	-	1	-	-	-
	BOOKINGID	NUMBER	22	-	-	-	✓	-	-
	AMOUNT	NUMBER	-	8	2	-	✓	-	-
	PAYMENTDATE	DATE	7	-	-	-	✓	-	-

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SQL Commands

Schema WKSP_DATABASE76

Language SQL

Rows 10

Clear Command

Find Tables

Save

Run

1 DESC Inventory;

2

Results

Explain

Describe

Saved SQL

History

Object Type TABLE

Object INVENTORY

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
INVENTORY	INVENTORYID	NUMBER	22	-	-	1	-	-	-
	ITEMNAME	VARCHAR2	100	-	-	-	✓	-	-
	QUANTITY	NUMBER	22	-	-	-	✓	-	-
	UNITPRICE	NUMBER	-	8	2	-	✓	-	-
	BRANCHID	NUMBER	22	-	-	-	✓	-	-

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RD

SQL Commands

Schema WKSP_DATABASE76

Language SQL

Rows 10

Clear Command

Find Tables

Save

Run

1 DESC Service;

2

Results

Explain

Describe

Saved SQL

History

Object Type TABLE

Object SERVICE

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
SERVICE	SERVICEID	NUMBER	22	-	-	1	-	-	-
	SERVICENAME	VARCHAR2	100	-	-	-	✓	-	-
	PRICE	NUMBER	-	8	2	-	✓	-	-

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SQL Commands

Schema WKSP_DATABASE76

Language SQL

Rows 10

Clear Command

Find Tables

Save

Run

1 DESC Booking;

2

Results

Explain

Describe

Saved SQL

History

Object Type TABLE

Object BOOKING

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
BOOKING	BOOKINGID	NUMBER	22	-	-	1	-	-	-
	BOOKINGDATE	DATE	7	-	-	-	✓	-	-
	CUSTOMERID	NUMBER	22	-	-	-	✓	-	-
	VEHICLEID	NUMBER	22	-	-	-	✓	-	-

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Search

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SQL Commands

Schema WKSP_DATABASE76

Language SQL

Rows 10

Clear Command

Find Tables

Save

Run

1 DESC Vehicle;

2

Results

Explain

Describe

Saved SQL

History

Object Type TABLE

Object VEHICLE

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
VEHICLE	VEHICLEID	NUMBER	22	-	-	1	-	-	-
	PLATENUMBER	VARCHAR2	20	-	-	-	✓	-	-
	CUSTOMERID	NUMBER	22	-	-	-	✓	-	-

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SQL Commands

Schema WKSP_DATABASE76

Language SQL

Rows 10

Clear Command

Find Tables

Save

Run

1 DESC Customer;

2

Results

Explain

Describe

Saved SQL

History

Object Type TABLE

Object CUSTOMER

Table	Column	Data Type	Length	Precision	Scale	Primary Key	Nullable	Default	Comment
CUSTOMER	CUSTOMERID	NUMBER	22	-	-	1	-	-	-
	CUSTOMERNAME	VARCHAR2	100	-	-	-	✓	-	-
	PHONE	VARCHAR2	20	-	-	-	✓	-	-

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- DML

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SQL Commands Schema WKSP_DATABASEG76

Language SQL Rows 10 Clear Command Find Tables Save Run

1 SELECT * FROM Payment;

Results Explain Describe Saved SQL History

PAYMENTID	BOOKINGID	AMOUNT	PAYMENTDATE
1	1	400	1/11/2026

1 rows returned in 0.02 seconds Download

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SQL Commands Schema WKSP_DATABASEG76

Language SQL Rows 10 Clear Command Find Tables Save Run

1 SELECT * FROM Inventory;

Results Explain Describe Saved SQL History

INVENTORYID	ITEMNAME	QUANTITY	UNITPRICE	BRANCHID
1	Engine Oil	50	25	1
2	Brake Pad	30	80	1

2 rows returned in 0.02 seconds Download

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SQL Commands Schema WKSP_DATABASEG76

Language SQL Rows 10 Clear Command Find Tables Save Run

1 SELECT * FROM Service;

Results Explain Describe Saved SQL History

SERVICEID	SERVICENAME	PRICE
1	Oil Change	100
2	Brake Repair	300

2 rows returned in 0.01 seconds Download

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SQL Commands Schema WKSP_DATABASEG76

Language SQL Rows 10 Clear Command Find Tables Save Run

1 SELECT * FROM Booking;

Results Explain Describe Saved SQL History

BOOKINGID	BOOKINGDATE	CUSTOMERID	VEHICLEID
1	1/11/2026	1	1
2	1/11/2026	2	2

2 rows returned in 0.01 seconds Download

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SQL Commands Schema WKSP_DATABASEG76

Language SQL Rows 10 Clear Command Find Tables Save Run

1 SELECT * FROM Vehicle;

Results Explain Describe Saved SQL History

VEHICLEID	PLATENUMBER	CUSTOMERID
2	XYZ789	2
1	ABC123	1

2 rows returned in 0.01 seconds Download

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SQL Commands Schema WKSP_DATABASEG76

Language SQL Rows 10 Clear Command Find Tables Save Run

1 SELECT * FROM Customer;

Results Explain Describe Saved SQL History

CUSTOMERID	CUSTOMERNAME	PHONE
2	Sara	0198765432
1	Ali	0123456789

2 rows returned in 0.01 seconds Download

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- Sample Query Results

The screenshot displays the Oracle APEX SQL Workshop interface. At the top, the 'SQL Commands' tab is active, showing a SQL query that joins the 'Booking' and 'Service' tables. The query is as follows:

```
1 SELECT b.BookingID, SUM(s.Price) AS TotalServiceCost
2 FROM Booking_Service bs
3 JOIN Booking b ON bs.BookingID = b.BookingID
4 JOIN Service s ON bs.ServiceID = s.ServiceID
5 GROUP BY b.BookingID;
```

Below the query editor, the 'Results' tab is selected, displaying a table with the following data:

BOOKINGID	CUSTOMERNAME	PLATENUMBER
1	Ali	ABC123
2	Sara	XYZ789

At the bottom of the results section, it indicates '2 rows returned in 0.02 seconds' and provides a 'Download' link. The footer of the interface shows the user '23098850@siswa.um.edu.my' and the version 'Oracle APEX 24.2.11'.

Shows **multi-table JOIN works** and fulfills a system requirement.

11. Discussion

During the development of the Raypro Tinted Expert database system, the group faced several challenges that helped improve our understanding of database design and implementation. One of the main problems was designing tables that could handle real business situations, such as assigning multiple staff members to one booking, allowing staff to have more than one role, and linking multiple services to a single booking. At first, it was difficult to decide how to structure these relationships without repeating data.

Another challenge was dealing with many-to-many relationships. In the early stages, we considered placing multiple values in one table, but this caused normalization issues. To solve this, we used bridge tables such as `STAFF_ROLE`, `BOOKING_SERVICE`, and `JOB_ASSIGNMENT`, which helped keep the database organized and in Third Normal Form (3NF).

The group also faced problems related to foreign key constraints, especially when creating tables or inserting data in the wrong order. Errors occurred when parent tables were not created before child tables. This taught us the importance of understanding table dependencies and planning the implementation process carefully.

From this experience, we learned that proper planning and normalization are essential when designing a database. We also learned that testing the database with sample data is important to ensure relationships and constraints work correctly. Overall, this project helped us gain practical experience in translating business requirements into a functional database system and improved our confidence in using SQL and relational database concepts.